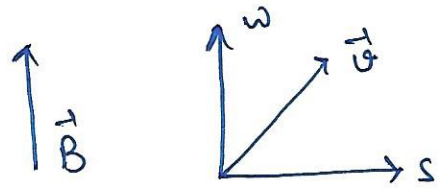
1. The speed of a point on the disk at a distance 's' from the axis is $|\vec{v}| = |\vec{\omega} \times \vec{s}| = \omega s$. Here '$\omega$' act upwards since direction of rotation is counterclockwise.

The force per unit charge, $\vec{f}_{mag} = \vec{v} \times \vec{B} = \omega s B \hat{s}$

The magnetic force is always along the radial direction.



(s is also along the radius of the disk)

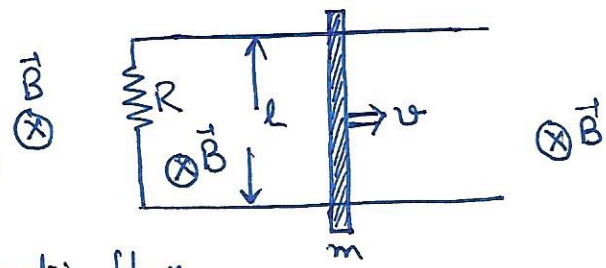
The induced emf is therefore

$$\mathcal{E} = \int_0^a f_{mag}\, ds = \omega B \int_0^a s\, ds$$

$$= \frac{\omega B a^2}{2}$$

The current is, $I = \frac{\mathcal{E}}{R} = \frac{\omega B a^2}{2R}$

2. The induced emf is given by the rate of change of magnetic flux (with a 'minus' sign)



$$\mathcal{E} = -\frac{d\phi}{dt}$$ [ϕ = magnetic flux enclosed by the region within resistor R and the bar]

$$= -Bl\frac{dx}{dt}$$

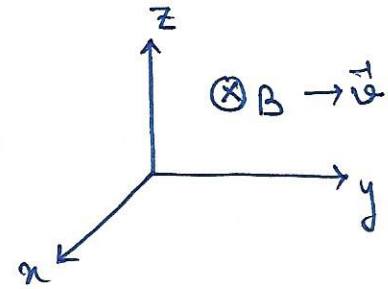
$$= -Blv$$. The induced current through the resistor is

$$\boxed{I = \frac{\mathcal{E}}{R} = \frac{Blv}{R}}$$ [− sign just tells about direction of current flow]

The positive charges in the metal bar experience a magnetic force (per unit charge) $= \vec{v} \times \vec{B}$ which is upwards in the bar and downward through the resistor, ie current flows in counterclockwise sense.

$\vec{f}_{mag} = \vec{v} \times \vec{B}$

$= v(\hat{y}) \times B(-\hat{x}) = vB(\hat{z})$ <u>in the metal bar</u>



(b) Due to current flow in the metal bar force acting is $\vec{F} = I(\vec{l} \times \vec{B}) = Il(\hat{z}) \times B(-\hat{x})$

$= IlB(-\hat{y})$ (towards the left)

$= \frac{Blv}{R} lB = \frac{B^2 l^2 v}{R}$ (towards left)

(c) Initial velocity $= v_0$ and the bar is left to slide

$$F = -m\frac{dv}{dt} = -\frac{B^2 l^2 v}{R} \Rightarrow \frac{dv}{dt} = -\frac{B^2 l^2 v}{mR}$$

$\frac{dv}{v} = -\frac{B^2 l^2}{mR} dt$. Integrating, $\ln v = -\frac{B^2 l^2}{mR} t + k$

Initial condition is: at $t = 0$, $v = v_0$

$$k = \ln v_0$$

$$\therefore \ln\left(\frac{v}{v_0}\right) = -\frac{B^2 l^2}{mR} t \Rightarrow \boxed{v = v_0 e^{-\frac{B^2 l^2}{mR} \times t}}$$

(d) Power delivered to the resistor, $\frac{dW}{dt} = I^2 R$

$$= \frac{B^2 l^2 v^2}{R}$$

$$= \frac{B^2 l^2}{R} v_0^2 e^{-2\alpha t} \left[\alpha = \frac{B^2 l^2}{mR}\right]$$

$$\frac{dW}{dt} = \alpha m v_0^2 e^{-2\alpha t}$$

$$W = \int dW = \alpha m v_0^2 \int_0^\infty e^{-2\alpha t} dt = \alpha m v_0^2 \left.\frac{e^{-2\alpha t}}{2\alpha}\right|_\infty^0$$

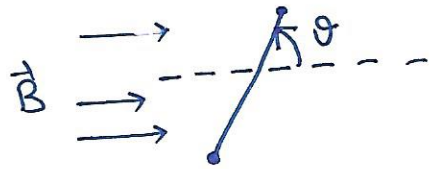
total energy delivered to the resistor.

$$= \frac{1}{2\alpha} \alpha m v_0^2 = \frac{1}{2} m v_0^2$$

3. We know flux of $\vec{B}$ through the closed loop is given by

$$\Phi = \vec{B}\cdot\vec{A} = BA\cos\theta \quad [A = \text{area of square loop}]$$

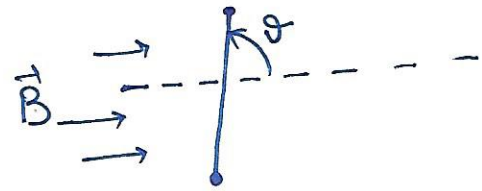
Due to angular rotation, the angle between area normal and the line of B axis is changing with time with angular velocity ω, that is $\theta = \omega t$. This is evident from top view



$$\mathcal{E} = -\frac{d\phi}{dt} = \text{induced emf}$$

$$= -\frac{d}{dt}(BA\cos\omega t) = BA\omega\sin\omega t$$

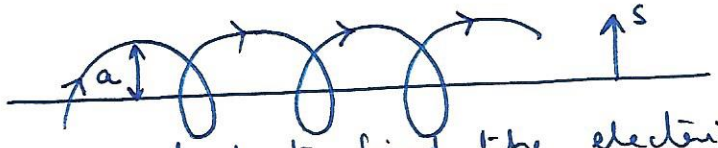
$$\boxed{\mathcal{E} = B\times a^2\times\omega\sin\omega t} \quad [A = a^2 = \text{area of square loop}]$$



(Top view)

This is the working principle of alternating current generators.

4.



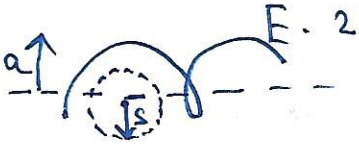
$\vec{I} = I(t)\,\hat{\phi}$. We are asked to find the electric field at a distance 's' from the axis. This electric field is an induced field (also called Faraday field) arising due to change in current I through the solenoid and I $\propto$ magnetic field.

We know $\oint \vec{E}\cdot d\vec{l} = \mathcal{E} = -\frac{d\phi}{dt}$ ——— (i)

induced (Faraday) field — induced emf

<u>Inside $s < a$</u> (apply eqn (i))

[$\vec{B}_{in} = \mu_0 n I\,\hat{z}$ for a solenoid, n = number of turns per unit length.]

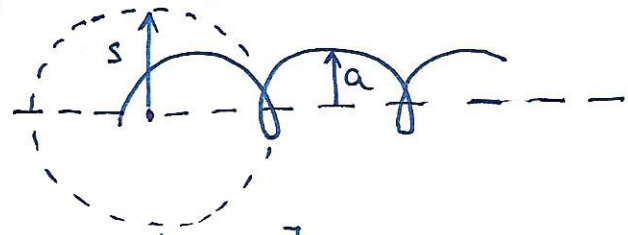


$$E\cdot 2\pi s = -\frac{d}{dt}(\mu_0 n I\times\pi s^2)$$

$$= -\mu_0 n\pi s^2\frac{dI}{dt}$$

$$\vec{E}_{in} = -\frac{\mu_0 n s}{2}\left(\frac{dI}{dt}\right)\hat{\phi}$$

<u>Outside $s > a$</u>. Consider an Amperian loop $s > a$



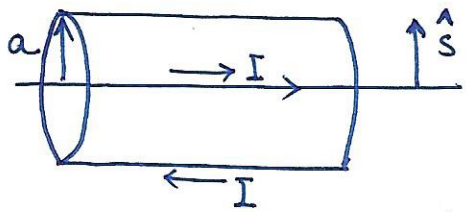
$$\oint \vec{E}\cdot d\vec{l} = -\frac{d}{dt}\phi$$

$$E \cdot 2\pi s = -\frac{d}{dt}(\pi a^2 \times \mu_0 n I)$$

[$B = 0$ outside solenoid]

$$= \mu_0 n \pi a^2 \left(\frac{dI}{dt}\right) \Rightarrow \boxed{\vec{E}_{out} = \frac{\mu_0 n a^2}{2s}\frac{dI}{dt}\hat{\phi}}$$

5. We are asked to find the induced electric field $\vec{E}(s,t)$

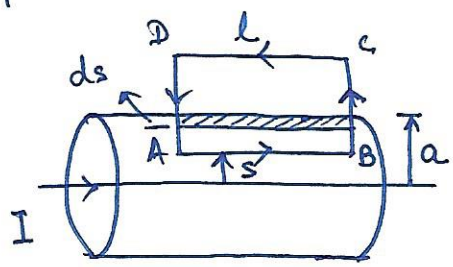


Here $I = I(t) = I_0 \cos\omega t$

It must be remembered that electric fields due to currents point along the direction of current (will be explained more in the class as electrodynamics lessons are progressed)

Note that outside the co-axial tube, $B = 0$ (due to contributions of $+I$ and $-I$) so the induced field $\vec{E}$ is also zero – after all induced field results from change in magnetic flux.

Consider a rectangular shaped Amperian loop as shown (because $\vec{E}$ is longitudinal) – apply $\oint \vec{E}\cdot d\vec{l} = -\frac{d\phi}{dt}$. Only segment AB will contribute



$$E \cdot l = -\frac{d\phi}{dt}$$. Magnetic field at a distance 's' is $\frac{\mu_0 I}{2\pi s}$ $\therefore E \cdot l = -\frac{d}{dt}\int \frac{\mu_0 I}{2\pi s} l\, ds$

total flux $\phi = \int \vec{B}\cdot d\vec{a}$ through closed loop.

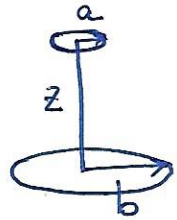
$$E\ell = -\frac{d}{dt}\int \frac{\mu_0 I}{2\pi}\ell \int_s^a \frac{ds}{s} = -\frac{\mu_0}{2\pi}\ell\frac{dI}{dt}\ln\left(\frac{a}{s}\right)$$

$$\vec{E} = -\frac{\mu_0}{2\pi}\frac{dI}{dt}\ln\left(\frac{a}{s}\right)\hat{z} \qquad \text{But } \frac{dI}{dt} = -I_0\omega\sin\omega t$$

$$\boxed{\vec{E} = \frac{\mu_0 I_0 \omega}{2\pi}\sin\omega t\,\ln\left(\frac{a}{s}\right)\hat{z}}$$

6.

(a) If current I flows in big loop, the magnetic field due to this circular current-carrying loop is



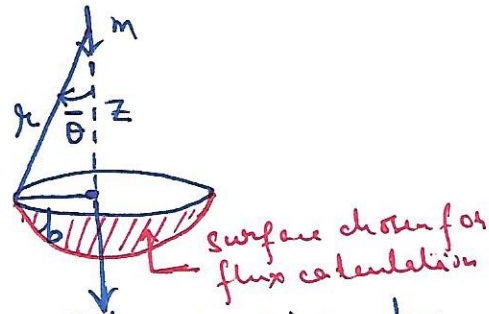
$$B(z) = \frac{\mu_0}{4\pi}\frac{2\pi I b^2}{(b^2+z^2)^{3/2}}$$ (Biot-Savart law, done in class)

Flux through loop a $= B\times\pi a^2 = \dfrac{\mu_0\pi I a^2 b^2}{2(b^2+z^2)^{3/2}} = \Phi_{ab}$

(b) Now consider that current loop (a) is so small that it behaves like a dipole - magnetic field due to dipole is

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{m}{r^3}\left[2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}\right]$$

This is true when $\vec{m}$ is aligned along z axis — $\hat{r}$ is the radial unit vector, θ is measured w.r.t z axis.



Now we can calculate the flux through bigger loop (b) by considering a "spherical cap" with the bigger loop as the boundary. Flux is given by $\Phi_{ba} = \int \vec{B}\cdot\vec{da}$. Note that we could have taken the cap to be of any arbitrary shape, for e.g flat surface [$\nabla\cdot B = 0 \Rightarrow \int \vec{B}\cdot\vec{da}$ to be independent of any surface for a <u>given boundary line</u>] - but spherical surface

is the most simplest surface. We can take the area element on the surface as $\vec{da} = r^2 \sin\theta \, d\theta \, d\varphi \, \hat{r}$

$$\Phi_{ba} = \int \vec{B} \cdot \vec{da} = \frac{\mu_0}{4\pi} \frac{I(\pi a^2)}{r^3} \int (2\cos\theta) \, r^2 \sin\theta \, d\theta \, d\varphi \quad [m = I \times \pi a^2]$$

$$= \frac{\mu_0}{2\pi} \frac{I(\pi a^2)}{r} \int_{\theta=0}^{\bar{\theta}} \cos\theta \sin\theta \, d\theta \int_0^{2\pi} d\varphi$$

Here $r = \sqrt{b^2 + z^2}$

$\sin\bar{\theta} = \frac{b}{r}$

$$\cancel{\Phi = \frac{\mu_0}{4\pi} \frac{I \pi a^2}{r} \cdot \frac{1}{2} \sin 2\bar{\theta}}$$

$$\Phi_{ba} = \frac{\mu_0}{2\pi} \frac{\pi I a^2}{r} \int_0^{\bar{\theta}} \sin\theta \, d(\sin\theta) \times 2\pi$$

$$= \mu_0 \pi I a^2 \left. \frac{\sin^2\theta}{2} \right|_0^{\bar{\theta}} = \frac{\mu_0 \pi I a^2 b^2}{2 r^3}$$

$$\Phi_{ba} = \frac{\mu_0 \pi I a^2 b^2}{2(b^2 + z^2)^{3/2}} = \Phi_{ab}$$

(c) Dividing Φ_{ba} or Φ_{ab} by I,

$$M = \frac{\mu_0 \pi a^2 b^2}{2(b^2 + z^2)^{3/2}}$$

7. Self inductance of a long solenoid of radius R. Number of

(a) turns per unit length = n

If the current I flows through each turn of solenoid, magnetic field $B = \mu_0 n I$ inside the solenoid

Flux through each turn of solenoid $= B \times \pi R^2$

$= \mu_0 n I \times \pi R^2$

In a length 'l', there are nl turns, so flux through the entire solenoid $\Phi_{tot} = \mu_0 n I \pi R^2 \times nl = \mu_0 n^2 R^2 \pi l$

$= LI$

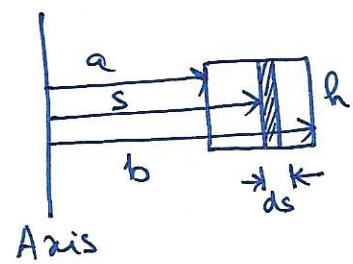
Self-inductance per unit length is

$$\boxed{\frac{L}{l} = \mu_0 n^2 \pi R^2}$$

(b) The magnetic field inside the toroid is

$$B = \frac{\mu_0 N I}{2\pi s}$$

s = radial distance from axis of toroid



Flux through a single turn is

$$\int \vec{B} \cdot d\vec{a} = \frac{\mu_0 N I}{2\pi} h \int_a^b \frac{ds}{s}$$

$$= \frac{\mu_0 N I h}{2\pi} \ln\left(\frac{b}{a}\right)$$

Total flux is N times this, so self inductance is

$$\boxed{L = \frac{\mu_0 N^2 h}{2\pi} \ln\left(\frac{b}{a}\right)}$$